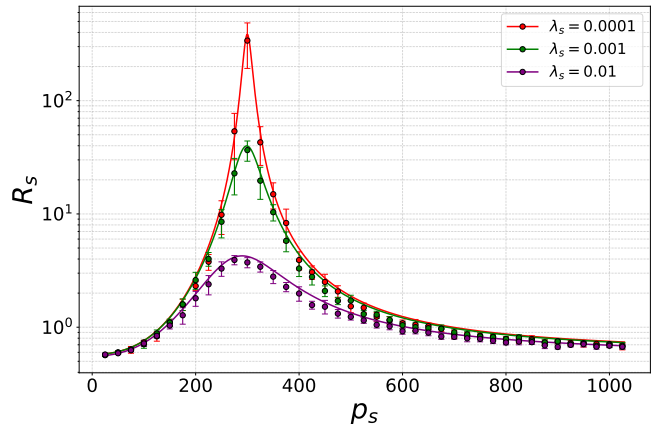
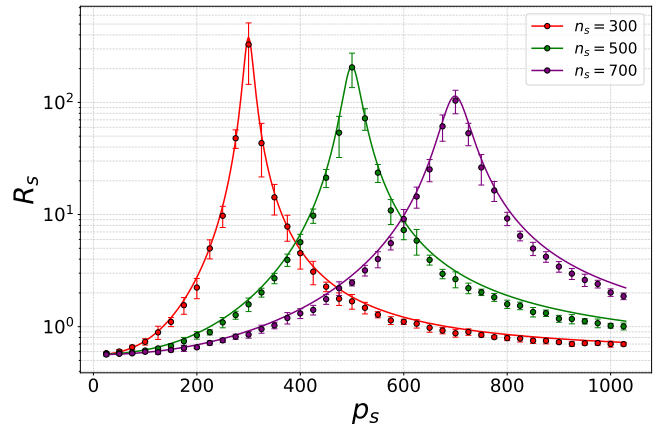
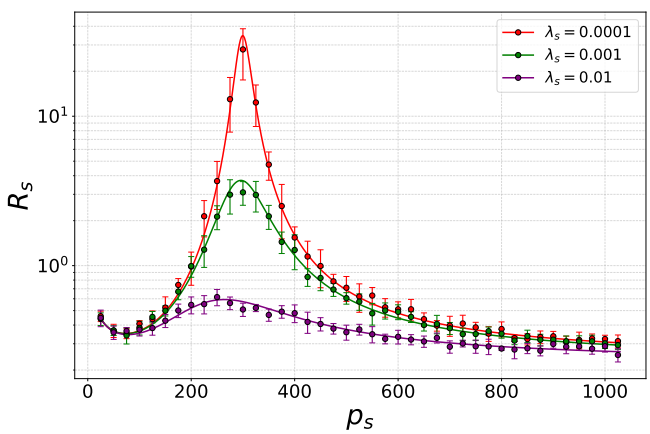


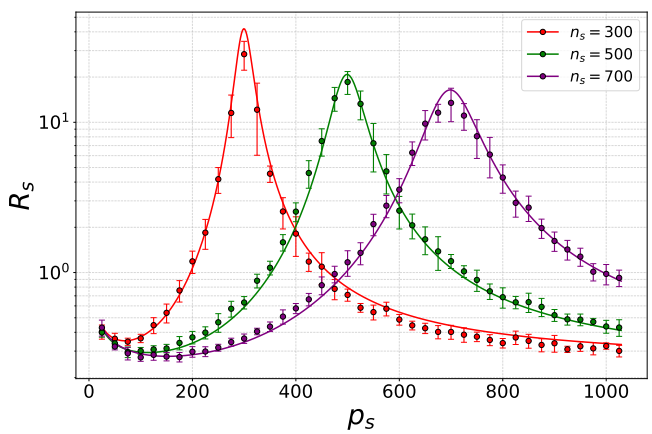
(a) Single-index target function, $n_\mathsf{s} = 300$.



(b) Single-index target function, $\lambda_\mathsf{s} = 10^{-4}$.

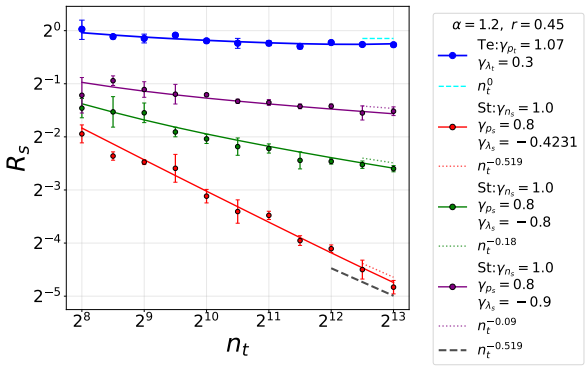


(c) MNIST data, $n_\mathsf{s} = 300$.

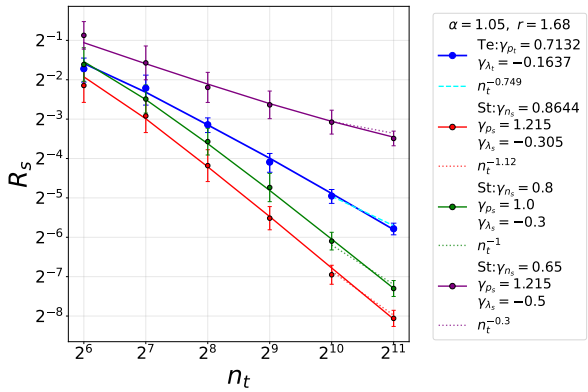


(d) MNIST data, $\lambda_\mathsf{s} = 10^{-4}$.

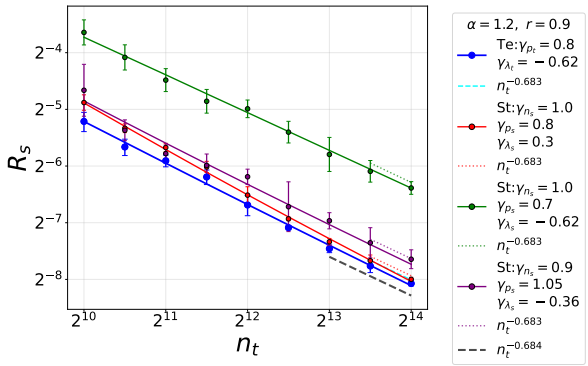
Figure 1: Excess test errors of various students trained on teacher labels, together with the corresponding deterministic equivalents (Theorem 2), as a function of the number of student features p_s. The teacher and the student are random feature models with the same number of features ($p_\mathsf{t} = p_\mathsf{s}$), trained with the same sample size ($n_\mathsf{t} = n_\mathsf{s}$) and with the teacher regularization being half of that of the student ($\lambda_\mathsf{t} = \lambda_\mathsf{s}/2$). In the top two plots, the data is given by $\boldsymbol{x}_i \sim \mathcal{N}(0, I_d)$ and $y_i = \mathrm{erf}(\langle \boldsymbol{x}_i, \boldsymbol{w}_* \rangle) + \varepsilon_i$, with $\boldsymbol{w}_* \sim \mathrm{Unif}(\mathbb{S}^{d-1})$, and the random feature model is $\varphi(\boldsymbol{x}; \boldsymbol{w}) = \tanh(\langle \boldsymbol{x}, \boldsymbol{w} \rangle)$, $\boldsymbol{w} \sim \mathrm{Unif}(\mathbb{S}^{d-1})$; in the bottom two plots, the data is obtained from the MNIST dataset and the random feature model is $\varphi(\boldsymbol{x}; \boldsymbol{w}) = \mathrm{erf}(\langle \boldsymbol{x}, \boldsymbol{w} \rangle)$, $\boldsymbol{w} \sim \mathrm{Unif}(\mathbb{S}^{d-1})$. In the left figures, we fix the sample size n_s and consider three values of the regularization λ_s (in three different colors); in the right figures, we fix the regularization λ_s and consider three values of the sample size n_s (in three different colors). We run 10 independent experiments, reporting the average and the confidence interval at 1 standard deviation. The circles represent the test errors obtained experimentally and they match well the continuous curves which represent the deterministic equivalents of Definition 2.



(a) Variance dominated: W2SG.



(b) Bias dominated: W2SG.



(c) Optimally tuned teacher: no W2SG.

Figure 2: Excess test errors of various students and teachers, as a function of the number of ground-truth samples n_{t}. We consider random feature ridge regression with weak-to-strong training, under the source and capacity conditions in (16) and with the hyperparameter scaling in (17). Each plot corresponds to a different choice of $(\alpha, r, \gamma_{p_{\mathrm{t}}}, \gamma_{\lambda_{\mathrm{t}}})$. Points are empirical experiments, solid lines are deterministic equivalent predictions (Theorems 1 and 2), dotted lines are theoretical decay rates (Theorems 3 and 4), and the grey dashed line is the minimax rate. The teacher test error is in blue, while the other colors correspond to student test errors for different choices of $(\gamma_{n_{\mathrm{s}}}, \gamma_{p_{\mathrm{s}}}, \gamma_{\lambda_{\mathrm{s}}})$.